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On the Lundgren hierarchy of helically symmetric turbulence

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Abstract

This paper analyzes the reduction of the infinite Lundgren–Monin–Novikov (LMN) hierarchy of probability density functions (PDFs) in the statistical theory of helically symmetric turbulence. Lundgren's hierarchy is considered a complete model, i.e. fully describes the joint multi-point statistic of turbulence though at the expense of dealing with an infinite set of integro-differential equations. The LMN hierarchy and its respective side-conditions are transformed to helical coordinates and thus are dimesionally reduced. In the course of development, a number of key questions were solved, namely in particular the transformation of PDFs and sample space velocities into orthonormal coordinate systems. In a validity check it is shown, that the mean momentum equations derived from the helical LMN hierarchy via statistical moment integration are identical to the mean momentum equations derived by direct ensemble averaging the Navier–Stokes equation, in helically symmetric form. Finally, we derive the equation for the characteristic function equivalent to the PDF equation in a helically symmetric frame, which allows to generate arbitrary n^{th}-order statistical moments by simple differentiation.

Keywords: LMN hierarchy, helically symmetric turbulence, probability density function

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1. Introduction

The development of a universal theory for turbulent flows is still an open research topic, despite the high practical relevance of turbulent flows. The statistical character of turbulent flows is undisputed, however, its description and computation based on statistical variables up to today remains a central open issue of classical physics. A characteristic of turbulent flows are eddies on a broad range of scales, which result from various causes in the flow, such as moving parts, boundary effects or instabilities. In view of helical symmetry in flows, certain technical devices, such as wind turbines, cyclones or delta wings cause the emergence of spirally symmetric, so-called helical structures (Vermeer *et al* 2003, Gupta and Kumar 2007). Further, helical vortex structures naturally arise as unstable modes as a result of vortex breakdown in swirling jets (Sarpkaya 1971). Helical instabilities in swirling flows occur not only in laminar, but also in turbulent flows.

Helically symmetric flows are of particular interest because all quantities depend on only two spatial coordinates and therefore dimensionally reduce the underlying equations of motion. However, since there are still three independent velocity components in addition to the two independent variables, the corresponding turbulence is positioned between 2D and 3D and is often referred to as 2 1/2 D turbulence. Two cases are to be distinguished here. If a velocity component along the helical coordinate is present, vortex stretching occurs and small scales may be generated—similar to 3D turbulence. If the velocity along the helical coordinate vanishes, no vortex stretching occurs, similar to 2D turbulence. This distinction is also found in the associated conservation laws, e.g. conservation of kinetic energy, momentum and helicity, which clearly differ between the two cases and in both cases there are very different infinite families of conservation laws (Kelbin *et al* 2013). Helically symmetric turbulence denotes a specific type of turbulent flow where statistical properties remain invariant under helical transformations. It is therefore different to helical turbulence where the flow possesses helicity, a quantity that measures the degree of twistiness in fluid flow through the dot product of velocity and vorticity vectors. Though it was found that for helically invariant inviscid flows the conservation of helicity can be vastly generalized.

In the last few decades, several authors have carried out a considerable amount of research on helical flows. Delbende *et al* (2012), developed a direct numerical simulation (DNS) code aimed at solving the Navier–Stokes equations for incompressible flow with helical symmetry giving access to the dynamics of one or several helical vortices. Kelbin *et al* (2013), Kelbin (2015) derived the Navier–Stokes equation in helical coordinates and in particular derived conservation laws for helical flows. Dierkes and Oberlack (2017) extended this work by deriving conservation laws for helical flows with a time-dependent pitch, and further, in Dierkes *et al* (2020, 2021), derived both exact solutions of the Navier–Stokes equation for helically symmetric viscous flows and further developed a high-order discontinuous Galerkin solver for these types of flows. Deußen *et al* (2020) showed that for helical flows all velocity components are generally non-zero and therefore the phenomenon of vortex stretching is present in the helical coordinate system and generates small scales of turbulent motion.

In order to describe the complete statistics of an incompressible turbulent flow, the infinite Lundgren–Monin–Novikov (LMN) hierarchy, derived by Lundgren (1967), Monin (1967) and Novikov (1968) is a suitable choice. In their theoretical frameworks, the evolution equations for the probability density functions (PDFs) of turbulent observables such as the velocity are obtained from the equations of fluid motion. The infinite hierarchy describes a system of integro-differential equations for the velocity distribution functions and is derived from the definitions of the probability distribution functions and the Navier–Stokes equations only. If a finite sub-set of equations is considered, an unclosed hierarchy of equations remains, where

the closure problem stems from the viscous and non-local pressure terms of the Navier–Stokes equations. Ulinich and Lyubimov (1969) formulated the LMN hierarchy for relative coordinates. The closure problem of the LMN hierarchy (Reynolds 1895) has been treated for example by Sinai and Yakhot (1989) and Yakhot (1989, 1998). Tatsumi (2011) closed the LMN hierarchy for the multi-point velocity distributions of fluid turbulence by making use of the cross-independence closure hypothesis. He obtained the minimum deterministic set of equations as the equations for the one-point velocity distribution, the two-point velocity distribution and the two-point local velocity distribution. Tatsumi and Yoshimura (2004, 2007) dealt with inertial and local similarity of the velocity distributions of homogeneous isotropic turbulence and derived a closed set of equations for the one-point and two-point velocity distributions. Wacławczyk *et al* (2017) investigated symmetries of the LMN hierarchy, where the invariance of the LMN hierarchy under translations, scaling and transformations was discussed and symmetries for the LMN hierarchy were derived. Some of them correspond to the statistical symmetries of the hierarchy of the multipoint correlation equations.

The theoretical framework of PDFs in the context of the LMN hierarchy has been investigated and applied to several specific flow configurations. Kollmann (1990, 2019) analyzed the closure-problem for single-point PDFs, discussed properties of the closed and unclosed terms in the PDF equations and formulated closure approaches for the molecular-transport and the fluctuating pressure-gradient terms. He furthermore dealt with the application of PDF methods to combustion as well as supersonic and reactive turbulent flows. Friedrich *et al* (2012) treated the LMN hierarchy and PDFs by introducing conditional moments and considering external forces. It was shown that the infinite chain of equations can be formally closed by replacing the PDF of order $n+1$ by conditional averages. Pope (2000) also provided a comprehensive review of closure models and simulation methods. A recently updated, comprehensive and modern introduction to the most important fields of probability theory has been collected by Klenke (2020) and Borovkov (2013).

Pope (2000) found that kinetic equations can be taken as a good starting point for modeling when instead of the coupling to higher order PDFs the equations are formally truncated on the single-point level. Pope and Ching (1993) obtained an exact expression for the PDF of any quantity in a stationary process in terms of its conditional time derivatives and found how that these conditional expectations influence its shape independent of all physical processes. Friedrich and Peinke (1997) calculated joint probability distributions for velocity increments of different length scales and provided evidence that the conditional probability distribution obeys a Fokker–Planck equation. Renner *et al* (2001) found a new approach to characterize small-scale turbulence, which leads to a Fokker–Planck equation for the r-evolution of the PDF of the velocity increment. This equation for is completely determined by the drift and diffusion coefficient which can be estimated directly from the experimental data. They showed that the Fokker–Planck equation describes the measured PDFs correctly, including intermittency effects. Pope (1985, 2000) further established PDF equations in combination with conditional averages for the investigation of reactive flows, based on the observation that the reaction kinetics is local in space, and, hence does not require the formulation of a closure in the corresponding PDF equation. Pope (1976, 1981, 1991) derived a transport equation for the joint PDF for modeling turbulent reactive flows and presented closure models for terms involving fluctuating pressure and mixing and further reviewed turbulent combustion models and showed that PDF models overcome closure problems as a single modeled equation for the joint PDF of velocity, dissipation and composition provide closure.

In the present work, the special case of helical-symmetric turbulence is statistically described by means of PDFs. For this purpose, the infinite LMN hierarchy and the corresponding side conditions are transformed to helical coordinates. From this, the corresponding moment equations and the formulation of the hierarchy of characteristic functions is derived.

2. Helical framework and Navier–Stokes equations

In the following section, helical coordinates are introduced and basis vectors are derived. Additionally, the moment equations for the helical coordinate system are presented.

The helical coordinate system is given by the coordinates r, ξ, η, based on the cylindrical coordinates r, z, ϕ, where $\xi = az + b\phi$ and a and b are constants. By choosing the constants a, b, a specific helical frame is obtained. Based on Cartesian coordinates a complete helical coordinate system reads

$$r = \sqrt{x^2 + y^2}, \qquad \xi = az + b \arctan\left(\frac{y}{x}\right), \qquad \eta = \frac{1}{b}z - \frac{1}{a}\arctan\left(\frac{y}{x}\right), \tag{1}$$

implying that $a, b \neq 0$.

Note that helical coordinates by (r, ξ, η) are not globally orthogonal. It can be shown that although the coordinates r, ξ are orthogonal, there exists no third coordinate globally orthogonal to both r and ξ. This has been proven by Vachagina and Kadyirov (2014). Nevertheless, a third orthonormal basis vector $\boldsymbol{e}_\eta$ in helical coordinates is constructed locally using the helical basis vectors expressed in Cartesian basis vectors by

$$\boldsymbol{e_r} = \begin{pmatrix} \frac{x}{\sqrt{x^2+y^2}} \\ \frac{y}{\sqrt{x^2+y^2}} \\ 0 \end{pmatrix}, \boldsymbol{e}_\xi = \begin{pmatrix} \frac{-by}{x^2+y^2} \\ \frac{bx}{x^2+y^2} \\ a \end{pmatrix} \cdot B, \boldsymbol{e}_\eta = \boldsymbol{e_r} \times \boldsymbol{e}_\xi = \begin{pmatrix} ay \\ -ax \\ b \end{pmatrix} \cdot \frac{B}{\sqrt{x^2+y^2}}, \tag{2}$$

with the abbreviation

$$B = B(x, y) = \frac{\sqrt{x^2 + y^2}}{\sqrt{a^2(x^2 + y^2) + b^2}}. \tag{3}$$

The helical coordinate system is illustrated in figure 1.

Note that different definitions are used by Kelbin and Dierkes in order to construct the third basis vector. For helically symmetric turbulence, all physical quantities, such as velocity $\boldsymbol{u}$ and pressure p, as well as the PDFs, which we introduce later in section 3, are independent of the helical coordinate η, i.e. $\frac{\partial}{\partial \eta} = 0$ applies. Independence from the coordinate η leads to the physical quantities and the corresponding equations depending on the independent variables r, ξ, t only, which simplifies the construction of exact solutions. Subsequently, compared to fully three-dimensional flows, helically symmetric flows are dimensionally reduced with respect to the independent variable. Therefore and as noted above, we refer to these flows as $2\,1/2$ dimensional, as all three velocity components in the subsequent helically symmetric equations Navier–Stokes equation (5) are still present and

$$\tilde{\boldsymbol{u}} = \left(u^r, u^\xi, u^\eta\right)^T \tag{4}$$

holds. As mentioned above, in three dimensional flows vortex stretching leads to the energy cascade from large to small scales, which does not occur in plane two-dimensional flows. On

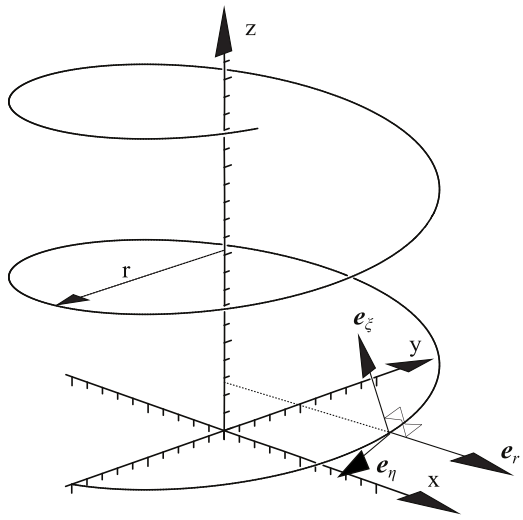

Figure 1. Helical coordinates with $\xi = const$. Kelbin *et al* (2013).

the contrary, in 2D the energy cascade is reversed. For helically symmetric flows the vortex stretching mechanism may still be active despite the dimensional reduction (Deußen *et al* 2020).

Using the definitions (1), the Navier–Stokes equations are transformed to a helical framework, reading

$$\frac{\partial u^r}{\partial r} + \frac{1}{r}u^r + \frac{1}{B}\frac{\partial u^\xi}{\partial \xi} = 0, \tag{5a}$$

$$\begin{aligned}\frac{\partial u^r}{\partial t} + u^r\frac{\partial u^r}{\partial r} + \frac{1}{B}u^\xi\frac{\partial u^r}{\partial \xi} - \frac{B^2}{r}\left(\frac{b}{r}u^\xi - au^\eta\right)^2 \\ = -\frac{\partial p}{\partial r} + \nu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u^r}{\partial r}\right) + \frac{1}{B^2}\frac{\partial^2 u^r}{\partial \xi^2} - \frac{1}{r^2}u^r + \frac{2bB}{r^2}\left(a\frac{\partial u^\eta}{\partial \xi} - \frac{b}{r}\frac{\partial u^\xi}{\partial \xi}\right)\right],\end{aligned} \tag{5b}$$

$$\begin{aligned}\frac{\partial u^\xi}{\partial t} + u^r\frac{\partial u^\xi}{\partial r} + \frac{1}{B}u^\xi\frac{\partial u^\xi}{\partial \xi} - \frac{2abB^2}{r}u^r u^\eta + \frac{b^2B^2}{r^2}u^r u^\xi \\ = -\frac{1}{B}\frac{\partial p}{\partial \xi} + \nu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u^\xi}{\partial r}\right) + \frac{1}{B^2}\frac{\partial^2 u^\xi}{\partial \xi^2} + \frac{a^4B^4 - 1}{r^2}u^\xi + \frac{2bB}{r}\left(\frac{b}{r^2}\frac{\partial u^r}{\partial \xi} - \frac{\partial \frac{aB}{r}u^\eta}{\partial r}\right)\right],\end{aligned} \tag{5c}$$

$$\begin{aligned}\frac{\partial u^\eta}{\partial t} + u^r\frac{\partial u^\eta}{\partial r} + \frac{1}{B}u^\xi\frac{\partial u^\eta}{\partial \xi} + \frac{a^2B^2}{r}u^r u^\eta \\ = \nu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u^\eta}{\partial r}\right) + \frac{1}{B^2}\frac{\partial^2 u^\eta}{\partial \xi^2} + \frac{a^2B^2\left(a^2B^2 - 2\right)}{r^2}u^\eta - \frac{2abB}{r^2}\left(\frac{\partial u^r}{\partial \xi} - \frac{\partial Bu^\xi}{\partial r}\right)\right],\end{aligned} \tag{5d}$$

where the constant density has been absorbed into the pressure p.

The range of the coordinates in order to span the entire space for r is the same as for cylindrical coordinates. For η the following applies:

$$\eta \in \left[\frac{\xi}{ab} - \frac{-\pi}{a}, \frac{\xi}{ab} - \frac{\pi}{a}\right]. \tag{6}$$

This becomes clear through the backwards transformation of the helical coordinates in (1) to Cartesian coordinates:

$$\begin{aligned} x &= \pm r \left| \cos\left(\frac{\xi - ab\eta}{2b}\right) \right| \\ y &= \pm r \left| \sin\left(\frac{\xi - ab\eta}{2b}\right) \right| \cdot sgn\left(\cos\left(\frac{\xi - ab\eta}{2b}\right)\right) \\ z &= \frac{1}{2}\left(\frac{\xi}{a} + b\eta\right), \end{aligned} \tag{7}$$

as the cylindrical coordinate $\phi = \frac{\xi - ab\eta}{2b}$ has the range of $\phi \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. Lastly $\xi \in [-\infty, \infty]$.

3. Multi-point PDFs and Lundgren's hierarchy

PDF methods provide a complete statistical description of turbulent flow fields at an arbitrary number of points. PDFs provide access to the mean values and all higher statistical moments of the flow quantities. As the LMN hierarchy is an infinite set of integro-differential equations for the PDFs, some basic properties of these should be explained first.

In the following, points in space are denoted by a bracketed prefixed index, i.e. $\boldsymbol{x}_{(1)}$ describes the location of the first point. In the statistical theory of turbulence, the velocity $\boldsymbol{u}$ is treated as a random variable and the average of any (Schwarz-)function Q, depending on the velocities, is expressed by

$$\langle Q\left(\boldsymbol{u}_{(1)}, \ldots, \boldsymbol{u}_{(n)}\right)\rangle = \int Q\left(\boldsymbol{v}_{(1)}, \ldots, \boldsymbol{v}_{(n)}\right) f\left(\boldsymbol{v}_{(1)}, \ldots, \boldsymbol{v}_{(n)}; \boldsymbol{x}_{(1)}, \ldots, \boldsymbol{x}_{(n)}\right) \mathrm{d}\boldsymbol{v}_{(1)} \ldots \mathrm{d}\boldsymbol{v}_{(n)}, \tag{8}$$

giving a weak definition of the PDF. Here $\boldsymbol{u}_{(i)} = \boldsymbol{u}(\boldsymbol{x}_{(i)}, t)$ denotes a realization of the velocity field, whereas $\boldsymbol{v}_{(i)}$ denotes the corresponding sample space velocity, which is an additional independent variable. Note that the PDF is a density with respect to $\boldsymbol{v}$ and a function with respect to the remaining variables $\boldsymbol{x}$ and t.

It is further possible to define velocity distributions using the ensemble average of a Dirac-delta function δ as the so-called fine-grained probability distribution function of a velocity at a given point. Thinking of the Dirac-delta function as the velocity distribution for one member of the ensemble, the PDF is the average of all distribution functions, leading to the definition of a PDF of nth-order by

$$f_{(n)} = f_{(n)}\left(\boldsymbol{v}_{(1)}, \ldots, \boldsymbol{v}_{(n)}; \boldsymbol{x}_{(1)}, \ldots, \boldsymbol{x}_{(n)}, t\right) = \left\langle \prod_{j=1}^{n} \delta\left(\boldsymbol{u}\left(\boldsymbol{x}_{(j)}, t\right) - \boldsymbol{v}_{(j)}\right) \right\rangle, \tag{9}$$

where $\langle \ldots \rangle$ denotes the ensemble average.

With the above definitions, the LMN hierarchy is derived from the incompressible Navier–Stokes equations (Lundgren 1967). The pressure p is eliminated using Green's function as a solution of Poisson's equation for an unbounded domain. In addition, all external sources are

neglected and viscosity ν and density ρ are assumed to be constant. Using the definition of the n-point PDF in (9), this results in the LMN hierarchy of nth-order.

$$\frac{\partial f_{(n)}}{\partial t} + \sum_{i=1}^{n} \underbrace{\boldsymbol{v}_{(i)} \cdot \nabla_{(i)} f_{(n)}}_{\mathcal{C}(\boldsymbol{v};\boldsymbol{x})} = -\sum_{j=1}^{n} \frac{\partial}{\partial \boldsymbol{v}_{(j)}} \cdot \Bigg[\underbrace{\lim_{\boldsymbol{x}_{(n+1)} \to \boldsymbol{x}_{(j)}} \nu \Delta_{(n+1)} \int \boldsymbol{v}_{(n+1)} f_{(n+1)} \mathrm{d}\boldsymbol{v}_{(n+1)}}_{\mathcal{V}(\boldsymbol{v};\boldsymbol{x})}$$
$$\underbrace{- \frac{1}{4\pi} \int \int \underbrace{\nabla_{(j)} (|\, \boldsymbol{x}_{(j)} - \boldsymbol{x}_{(n+1)} \,|)^{-1}}_{\mathcal{G}(\boldsymbol{x}_{(j)}, \boldsymbol{x}_{(n+1)})} (\boldsymbol{v}_{(n+1)} \cdot \nabla_{(n+1)})^2 f_{(n+1)} \mathrm{d}\boldsymbol{x}_{(n+1)} \mathrm{d}\boldsymbol{v}_{(n+1)}}_{\mathcal{P}(\boldsymbol{v};\boldsymbol{x})} \Bigg], \tag{10}$$

where $\mathcal{C}$ denotes the convective term, $\mathcal{V}$ the viscous term, $\mathcal{G}$ the gradient of Green's function and $\mathcal{P}$ the full pressure integral. As each equation in (10) introduces one higher-order distribution function, it represents an unclosed set of equations, if the hierarchy is truncated. Key properties of the hierarchy are linearity in all $f_{(i)}$ and non-locality in both $\boldsymbol{x}_{(i)}$ and $\boldsymbol{v}_{(i)}$.

Equation (10) shall now be transformed into helical coordinates. Therefore, position vectors, sample space velocities, derivatives, volume elements and PDFs need to be transformed accordingly.

The LMN hierarchy and the associated PDF have to comply with a number of side conditions, which must be fulfilled as well, and these will be transformed and discussed in detail in section 3.4.

3.1. Lundgren's hierarchy in helical coordinates

In the following, we will transform the LMN hierarchy to a helical coordinate system. In view of equation (1), the transformation can thus be traced back to Cartesian coordinates. We will show in section 4 that the 1st moment equations, derived from the transformed LMN hierarchy, match those derived from the Navier–Stokes equations. Figure 2 describes the procedure schematically.

We recall that Cartesian position vectors are denoted by $\boldsymbol{x}$, Cartesian PDFs by f and Cartesian sample space velocities by $\boldsymbol{v}$. The respective transformed elements in helical coordinates are denoted by a tilde, i.e.

$$\tilde{\boldsymbol{x}} = r \cdot \boldsymbol{e_r} + \xi \cdot \boldsymbol{e}_\xi + \eta \cdot \boldsymbol{e}_\eta, \quad \tilde{\boldsymbol{v}} = v^r \cdot \boldsymbol{e_r} + v^\xi \cdot \boldsymbol{e}_\xi + v^\eta \cdot \boldsymbol{e}_\eta \tag{11}$$

and $\tilde{f}$ is meant to depend on the latter variables. We further remind that, the physical quantities for helically symmetric flows are assumed to be independent of η. In view of the definition of the PDF in (9), it follows that $\tilde{f}$ is also independent from η.

3.2. PDFs in helical coordinates

In the following, we will point out the special properties of PDFs when transforming to a curvilinear system. The PDF is a scalar density, i.e. a tensor density of order zero and weight -1, whose integral yields an invariant. PDFs are not scalars and therefore not coordinate

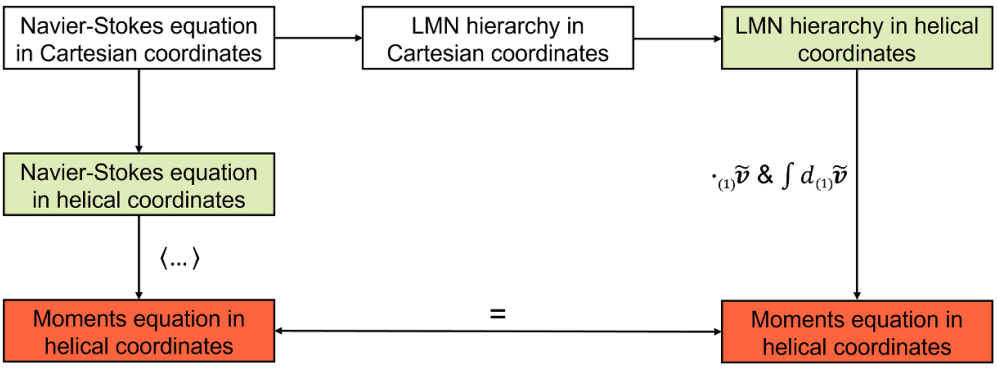

Figure 2. Alternative ways of deriving moment equations.

independent. Instead, the probability measure is the invariant under coordinate transformation, given by Risken (1996)

$$f_{(n)}\left(V,X\right)\mathrm{d}V=\tilde{f}_{(n)}\left(\tilde{V},\tilde{X}\right)\mathrm{d}\tilde{V}. \tag{12}$$

Here the set of sample space velocity vectors is denoted by

$$V=\boldsymbol{v}_{(1)},\ldots,\boldsymbol{v}_{(n)} \tag{13}$$

resp. $\tilde{V}$ in the new coordinate system and the collection of spatial positions analogous by

$$X=\boldsymbol{x}_{(1)},\ldots,\boldsymbol{x}_{(n)} \tag{14}$$

resp. $\tilde{X}$. The sample space velocity volume elements transform according to

$$\mathrm{d}\boldsymbol{v}_{(n)}=\mathrm{d}\tilde{\boldsymbol{v}}_{(n)}J_{(n)}, \qquad \mathrm{d}V=\mathrm{d}\tilde{V}J, \tag{15}$$

where $J_{(n)}$ denotes the Jacobian determinant of the transformation of the sample space velocity basis of $\boldsymbol{v}_{(n)}$ and $J=\prod_{k=1}^{n}J_{(k)}$. Together with (12), we observe that

$$Jf_{(n)}=\tilde{f}_{(n)}. \tag{16}$$

Presently, the basis vectors of the sample space variables are the basis vectors of the helical coordinate system (2). These are by definition normalized and orthogonal, which means that

$$J_{(k)}=1, \qquad J=1, \qquad \Rightarrow \qquad \nabla_{(i)}J_{(j)}=0, \qquad \frac{\partial}{\partial v_{(i)}}J_{(j)}=0. \tag{17}$$

We therefore conclude that

$$f=\tilde{f}, \quad \mathrm{d}\boldsymbol{v}=\mathrm{d}\tilde{\boldsymbol{v}}=\mathrm{d}v^{r}\mathrm{d}v^{\xi}\,\mathrm{d}v^{\eta} \quad \text{and} \quad \mathrm{d}V=\mathrm{d}\tilde{V} \tag{18}$$

is always true when transforming from Cartesian coordinates to an orthonormal curvilinear coordinate system.

3.3. Transformation rules

3.3.1. Derivatives and Volume elements. The transformation of the nabla-operator w.r.t. the sample space velocity yields

$$\frac{\partial}{\partial \boldsymbol{v}} = \frac{\partial}{\partial v^x}\boldsymbol{e_x} + \frac{\partial}{\partial v^y}\boldsymbol{e_y} + \frac{\partial}{\partial v^z}\boldsymbol{e_z} = \frac{\partial}{\partial v^r}\boldsymbol{e_r} + \frac{\partial}{\partial v^\xi}\boldsymbol{e}_\xi + \frac{\partial}{\partial v^\eta}\boldsymbol{e}_\eta = \frac{\partial}{\partial \tilde{\boldsymbol{v}}}, \tag{19}$$

where its derivation is given in appendix A.1.

As usual, the nabla-operator $\nabla_{(i)} = \partial_{(i)} = \frac{\partial}{\partial \boldsymbol{x}_{(i)}}$ is transformed using the Jacobian $\boldsymbol{K}$ of the transformation of the coordinates, where the abbreviation $\theta = -\frac{ab\eta - \xi}{2b}$ is used, yielding

$$\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) = \left(\frac{\partial}{\partial r}, \frac{\partial}{\partial \xi}, \frac{\partial}{\partial \eta}\right) \cdot \boldsymbol{K}^{-1}, \tag{20}$$

with

$$\boldsymbol{K} = \begin{bmatrix} \cos\theta & -\frac{r\sin\theta}{2b} & \frac{ar\sin\theta}{2} \\ \sin\theta & \frac{r\cos\theta}{2b} & -\frac{ar\cos\theta}{2} \\ 0 & \frac{1}{2a} & \frac{b}{2} \end{bmatrix}. \tag{21}$$

The realization of the nabla operator in helical coordinates then reads

$$\tilde{\nabla} = \boldsymbol{e_r}\frac{\partial}{\partial r} + \frac{1}{B}\boldsymbol{e}_\xi\frac{\partial}{\partial \xi} + \frac{B\left(a^2r^2 - b^2\right)}{bar^2}\boldsymbol{e}_\xi\frac{\partial}{\partial \eta} + \frac{2B}{r}\boldsymbol{e}_\eta\frac{\partial}{\partial \eta}. \tag{22}$$

With $\frac{\partial}{\partial \eta} = 0$ we find

$$\tilde{\nabla} = \boldsymbol{e_r}\frac{\partial}{\partial r} + \frac{1}{B}\boldsymbol{e}_\xi\frac{\partial}{\partial \xi}. \tag{23}$$

Deriving the Laplace-operator from (22) by dual application of the nabla-operator results in

$$\begin{aligned}\tilde{\Delta} = \tilde{\nabla}^2 = &\frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{B^2}\frac{\partial^2}{\partial \xi^2} + \frac{2\left(a^2r^2 - b^2\right)}{bar^2}\frac{\partial^2}{\partial \xi\,\partial \eta} + \frac{4B^2}{r^2}\frac{\partial^2}{\partial \eta^2} \\ &+ \left(\frac{B\left(a^2r^2 - b^2\right)}{abr^2}\right)^2\frac{\partial^2}{\partial \eta^2}.\end{aligned} \tag{24}$$

Applying $\frac{\partial}{\partial \eta} = 0$ gives in case of helical symmetry quantities the Laplacian

$$\tilde{\Delta} = \frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{B^2}\frac{\partial^2}{\partial \xi^2}. \tag{25}$$

The Jacobian determinant $det(\boldsymbol{K}) = \frac{r}{2}$ is used to transform the volume element $\mathrm{d}\boldsymbol{x} = \mathrm{d}x\mathrm{d}y\mathrm{d}z$. The entire $\mathbb{R}^3$ is spanned by $\eta \in \left[\frac{\xi}{(ab)} - \frac{2}{a}, \frac{\xi}{(ab)} - \frac{2\pi}{a}\right]$, $\xi \in [-\infty, \infty]$ and $r \in [0, \infty]$. This is also evident from figure 1. The volume element in helical coordinates reads

$$\mathrm{d}\boldsymbol{x} = det\left(\boldsymbol{K}\right)\mathrm{d}r\mathrm{d}\xi\,\mathrm{d}\eta = \frac{r}{2}\mathrm{d}\tilde{\boldsymbol{x}}. \tag{26}$$

The determinant of $\boldsymbol{K}$ differs from J since the third basis vector of the sample space velocity basis is derived according to (2) as the cross product of the first two basis vectors and therefore locally orthogonal, while $\boldsymbol{K}$ results from the derivatives of the coordinate lines. We further note that subsequently, e.g. in equation (32), the differential $\mathrm{d}\eta$ disappears as the PDF is assumed to be independent of η and hence any explicit η-appearance can be integrated out.

In the following, we will make use of the acquired knowledge to subsequently transform all terms occurring in the LMN hierarchy (10).

3.3.2. Convective term $\mathcal{C}$. As the sample space velocity is considered as a vector field, we may make use of $\nabla \cdot \boldsymbol{v} = 0$. In Cartesian coordinates $\boldsymbol{v}$ is independent from $\boldsymbol{x}$ and thus its divergence as a scalar invariant under coordinate transformations. In helical coordinates the convective term reads

$$\mathcal{C}\left(\tilde{V};\tilde{X}\right) = \frac{\partial f_{(n)}\boldsymbol{v}_{(i)}}{\partial \boldsymbol{x}_{(i)}} = v^{r}_{(i)}\frac{\partial \tilde{f}_{(n)}}{\partial r_{(i)}} + \frac{v^{\xi}_{(i)}}{B_{(i)}}\frac{\partial \tilde{f}_{(n)}}{\partial \xi_{(i)}}, \tag{27}$$

where we have already assumed a helical symmetric flow, i.e. $\frac{\partial}{\partial \eta} = 0$.

3.3.3. Viscous term $\mathcal{V}$. For the transformation of the viscous term, the Laplacian needs to be transformed. One can achieve this by applying the nabla-operator twice (see appendix A.2). Applying independency of η results in

$$\begin{aligned}\mathcal{V}\left(\tilde{V};\tilde{X}\right) = \frac{\partial}{\partial \tilde{\boldsymbol{v}}_{(j)}}\Bigg(&\lim_{\tilde{\boldsymbol{x}}_{(n+1)}\to\tilde{\boldsymbol{x}}_{(j)}} \nu\left[\frac{\partial^2}{\partial {r_{(n+1)}}^2} + \frac{1}{{B_{(n+1)}}^2}\frac{\partial^2}{\partial {\xi_{(n+1)}}^2} + \frac{1}{r_{(n+1)}}\frac{\partial}{\partial r_{(n+1)}}\right] \\ &\times \int \tilde{\boldsymbol{v}}_{(n+1)}\tilde{f}_{(n+1)}\mathrm{d}\tilde{\boldsymbol{v}}_{(n+1)}\Bigg),\end{aligned} \tag{28}$$

with $1 \leqslant j \leqslant n$.

3.3.4. Green's term $\mathcal{G}$. We call $\mathcal{G}$, as it is defined in (10), 'Green's term', since it is derived from Green's function as a solution of the Poisson equation. In order to transform this term we rewrite the integral using increments r_I, ξ_I, η_I. Note that for concrete calculations of differences between vectors, it is necessary to express everything w.r.t. a Cartesian basis, as otherwise the basis vectors at two different points point in different directions. Therefore, we express the increments w.r.t. a Cartesian basis, meaning that $\boldsymbol{x}_I = -(\boldsymbol{x}_{(j)} - \boldsymbol{x}_{(n+1)})$. This allows to calculate the inner integral of the pressure term $\mathcal{P}$ in (10), assuming the pdf being independent of η_I in case of helical symmetric turbulence. The inner integral reads with the correct limits of integration from (6)

$$\begin{aligned}\mathcal{G}'(\tilde{\boldsymbol{x}}_{(j)},\tilde{\boldsymbol{x}}_{(n+1)}) &= \int_{\frac{\xi_{(n+1)}+\pi b}{ab}}^{\frac{\xi_{(n+1)}-\pi b}{ab}} \mathcal{G}(\tilde{\boldsymbol{x}}_{(j)},\tilde{\boldsymbol{x}}_{(n+1)})\mathrm{d}\eta_{(n+1)} \\ &= \frac{\partial}{\partial \boldsymbol{x}_{(j)}}\int_{\frac{\xi_{(n+1)}+\pi b}{ab}}^{\frac{\xi_{(n+1)}-\pi b}{ab}} \left(|\boldsymbol{x}_{(j)} - \boldsymbol{x}_{(n+1)}|\right)^{-1}\mathrm{d}\eta_{(n+1)}\end{aligned}$$

$$= \frac{\partial}{\partial \boldsymbol{x}_{(j)}} \int_{\frac{\xi_{(n+1)}+\pi b}{ab}}^{\frac{\xi_{(n+1)}-\pi b}{ab}} \frac{2}{\sqrt{4r_I^2 + (\frac{\xi_I}{a} + b\eta_I)^2}} \mathrm{d}\eta_I$$

$$= \frac{\partial}{\partial \boldsymbol{x}_{(j)}} \frac{2}{b} \ln\left(\frac{\frac{b\xi_I}{a} + b^2\eta_I}{b} + \sqrt{\eta_I^2 b^2 + \frac{2b\xi_I\eta_I}{a} + \frac{4r_I^2a^2 + \xi_I^2}{a^2}} \right) \Bigg|_{\frac{\xi_{(n+1)}+\pi b}{ab}}^{\frac{\xi_{(n+1)}-\pi b}{ab}} \tag{29}$$

Inserting the bounds of integration brings us to

$$\mathcal{G}' = \tilde{\nabla}_{(j)} \left(\frac{2}{b} \ln\left[\frac{2b}{a}(1-\pi) + \frac{2}{a}\left(\sqrt{(\xi_I - \pi b)^2 + r_I^2 a^2} - \sqrt{(\xi_I - b)^2 + r_I^2 a^2} \right) \right] \right). \tag{30}$$

We find that the coordinate η_I has vanished, meaning that we have formally obtained a reduction for the helical symmetric case. Further calculation steps are necessary to transform the latter expression back into absolute coordinates. These and corresponding approaches are quite technical and therefore shown in appendix A.3.

3.3.5. Pressure term $\mathcal{P}$. The reduced Green's term is now inserted into the full pressure term in (10), yielding

$$\begin{aligned}\mathcal{P}(\tilde{\boldsymbol{v}}, \tilde{\boldsymbol{x}}) &= -\frac{1}{4\pi} \int \mathcal{G}' \left(\tilde{\boldsymbol{v}}_{(n+1)} \cdot \tilde{\nabla}_{(n+1)} \right)^2 \tilde{f}_{(n+1)} \mathrm{d}r_{(n+1)} \mathrm{d}\xi_{(n+1)} \mathrm{d}\tilde{\boldsymbol{v}}_{(n+1)} \\ &= -\frac{1}{4\pi} \int \mathcal{G}' \left(\tilde{\boldsymbol{v}}_{(n+1)} \cdot \left(\boldsymbol{e}_{r_{(n+1)}} \frac{\partial}{\partial r_{(n+1)}} + \frac{1}{B_{(n+1)}} \boldsymbol{e}_{\xi_{(n+1)}} \frac{\partial}{\partial \xi_{(n+1)}} \right) \right)^2 \\ &\quad \times \tilde{f}_{(n+1)} \frac{r_{(n+1)}}{2} \mathrm{d}r_{(n+1)} \mathrm{d}\xi_{(n+1)} \mathrm{d}\tilde{\boldsymbol{v}}_{(n+1)}.\end{aligned} \tag{31}$$

Assembling the transformed term (31), (28) and (27) into (10), the reduced LMN hierarchy of nth-order is given by

$$\begin{aligned}&\left[\frac{\partial}{\partial t} + \sum_{i=1}^{n} v_{(i)}^r \cdot \frac{\partial}{\partial r_{(i)}} + v_{(i)}^{\xi} \frac{1}{B_{(i)}} \cdot \frac{\partial}{\partial \xi_{(i)}} \right] \tilde{f}_{(n)} \\ &= -\sum_{j=1}^{n} \frac{\partial}{\partial \tilde{\boldsymbol{v}}_{(j)}} \cdot \Bigg\{ -\frac{1}{4\pi} \int \left[\tilde{\boldsymbol{v}}_{(n+1)} \cdot \left(\boldsymbol{e}_{r_{(n+1)}} \frac{\partial}{\partial r_{(n+1)}} + \frac{1}{B_{(n+1)}} \boldsymbol{e}_{\xi_{(n+1)}} \frac{\partial}{\partial \xi_{(n+1)}} \right) \right]^2 \\ &\quad \times \tilde{f}_{(n+1)} \mathcal{G}'\left(\tilde{\boldsymbol{x}}_{(j)}, \tilde{\boldsymbol{x}}_{(n+1)} \right) \frac{r_{(n+1)}}{2} \mathrm{d}r_{(n+1)} \mathrm{d}\xi_{(n+1)} \mathrm{d}\tilde{\boldsymbol{v}}_{(n+1)} \\ &\quad + \lim_{\tilde{\boldsymbol{x}}_{(n+1)} \to \tilde{\boldsymbol{x}}_{(j)}} \nu \left[\frac{\partial^2}{\partial r_{(n+1)}^2} + \frac{1}{r_{(n+1)}} \frac{\partial}{\partial r_{(n+1)}} + \frac{1}{B_{(n+1)}^2} \frac{\partial^2}{\partial \xi_{(n+1)}^2} \right] \int \tilde{\boldsymbol{v}}_{(n+1)} \tilde{f}_{(n+1)} \mathrm{d}\tilde{\boldsymbol{v}}_{(n+1)} \Bigg\},\end{aligned} \tag{32}$$

where $\tilde{\boldsymbol{v}}$ and $\mathrm{d}\tilde{\boldsymbol{v}}$ are respectively defined in equations (11) and (18) and $\mathcal{G}'$ may be taken from (30). As was intended from the outset, we find that the hierarchy for helically symmetric turbulence (32) is independent of η and therefore dimensionally reduced.

3.4. Transformation of the PDF's side conditions

PDFs in general have to fulfill several side conditions due to physical and mathematical conditions (Lundgren 1967). In the following, these conditions are discussed for the case of helically symmetric turbulence.

3.4.1. Normalization and reduction conditions. Both the normalization and the reduction conditions follow from the mathematical properties holding for any PDF. The normalization property (33) is a special case of the reduction property (34) (Lundgren 1967, Friedrich *et al* 2012, Staffolani *et al* 2014), i.e.

$$\int f_{(n)} \mathrm{d}V_{(n)} = 1, \tag{33}$$

where the integrals refer to an interpretation over all sample space velocities $\boldsymbol{v}_{(1)} \dots \boldsymbol{v}_{(n)}$ as defined in (13) and

$$\int f_{(n)} \mathrm{d}\boldsymbol{v}_{(n)} = f_{(n-1)}. \tag{34}$$

Inserting the transformed PDF and sample space velocity volume (18) into the normalization condition yields

$$\int \tilde{f}_{(n)} \mathrm{d}\tilde{V}_{(n)} = 1, \tag{35}$$

and the transformed reduction condition reads

$$\int \tilde{f}_{(n+1)} \mathrm{d}\tilde{\boldsymbol{v}}_{(n+1)} = \tilde{f}_{(n)}. \tag{36}$$

3.4.2. Coincidence condition. The velocities at two coinciding points must be the same, emerging from the fact that the velocity field is assumed to be smooth, resulting in the so called coincidence condition

$$\lim_{\boldsymbol{x}_{(n+1)} \to \boldsymbol{x}_{(n)}} f_{(n+1)} = f_{(n)} \delta\left(\boldsymbol{v}_{(n+1)} - \boldsymbol{v}_{(n)}\right). \tag{37}$$

Inserting the transformed PDF (18) yields

$$\lim_{\tilde{\boldsymbol{x}}_{(n+1)} \to \tilde{\boldsymbol{x}}_{(n)}} \tilde{f}_{(n+1)} = \tilde{f}_{(n)} \delta\left(\boldsymbol{v}_{(n+1)} - \boldsymbol{v}_{(n)}\right). \tag{38}$$

Note that in general the delta distribution needs to be transformed as well when transforming its argument to a curvilinear system, stemming from the fact that for any function Q

$$Q\left(\boldsymbol{v}_{(n)}\right) = \int \delta\left(\boldsymbol{v}_{(n+1)} - \boldsymbol{v}_{(n)}\right) Q\left(\boldsymbol{v}_{(n+1)}\right) \mathrm{d}\boldsymbol{v}_{(n+1)} \overset{!}{=} Q\left(\tilde{\boldsymbol{v}}_{(\boldsymbol{n})}\right), \tag{39}$$

since $\tilde{\boldsymbol{v}}_{(\boldsymbol{n})}$ and $\boldsymbol{v}_{(n)}$ represent the same vector in different systems. Transforming the volume element on the left hand side and using the definition of the delta distribution in curvilinear coordinates on the right hand side of the latter equation brings us to

$$\int \delta\left(\boldsymbol{v}_{(n+1)} - \boldsymbol{v}_{(n)}\right) Q\left(\tilde{\boldsymbol{v}}_{(n+1)}\right) J_{(n+1)} \mathrm{d}\tilde{\boldsymbol{v}}_{(n+1)} \overset{!}{=} \int \delta\left(\tilde{\boldsymbol{v}}_{(n+1)} - \tilde{\boldsymbol{v}}_{(n)}\right) Q\left(\tilde{\boldsymbol{v}}_{(n+1)}\right) \mathrm{d}\tilde{\boldsymbol{v}}_{(n+1)}. \tag{40}$$

From this, it directly follows that

$$\delta\left(\boldsymbol{v}_{(n+1)}-\boldsymbol{v}_{(n)}\right)J_{(n+1)}=\delta\left(\tilde{\boldsymbol{v}}_{(n+1)}-\tilde{\boldsymbol{v}}_{(n)}\right). \tag{41}$$

Recalling that in the present system as shown in (17), all Jacobian determinants of the sample space velocity basis transformations are equal one, we obtain

$$\delta\left(\boldsymbol{v}_{(n+1)}-\boldsymbol{v}_{(n)}\right)=\delta\left(\tilde{\boldsymbol{v}}_{(n+1)}-\tilde{\boldsymbol{v}}_{(n)}\right), \tag{42}$$

Note that in general the sample space velocity vectors in the argument of the delta distribution have to be expressed with respect to the local basis for concrete evaluation, which is different from the 'natural' expression of the sample space velocity vectors at nth and $(n+1)$th point with respect to the local basis at these points. Equation (38) then yields

$$\lim_{\tilde{\boldsymbol{x}}_{(n+1)}\to\tilde{\boldsymbol{x}}_{(n)}}\tilde{f}_{(n+1)}=\tilde{f}_{(n)}\delta\left(\tilde{\boldsymbol{v}}_{(n+1)}-\tilde{\boldsymbol{v}}_{(n)}\right). \tag{43}$$

3.4.3. Divergence condition. Since an incompressible flow is considered, the averaged velocity field is divergence-free and the divergence condition represents the nth order 'continuity' equation (Lundgren 1967, Hosokawa 2006, Friedrich *et al* 2012, Staffolani *et al* 2014). Directly derived from $\nabla\cdot\boldsymbol{u}=0$, the divergence condition reads

$$\nabla_{(i)}\cdot\int\boldsymbol{v}_{(i)}f_{(n)}\mathrm{d}\boldsymbol{v}_{(i)}=0,\quad\forall i\in\{1,\ldots,n\}. \tag{44}$$

Inserting the transformations (18) and (22) leads to

$$\int\tilde{\boldsymbol{v}}_{(i)}\cdot\tilde{\nabla}\tilde{f}_{(n)}\mathrm{d}\tilde{\boldsymbol{v}}_{(i)}=0.\quad\text{with}\quad i\leqslant n \tag{45}$$

with the definitions $\tilde{\boldsymbol{v}}_{(i)}$, $\mathrm{d}\tilde{\boldsymbol{v}}_{(i)}$ and $\tilde{\nabla}_{(i)}$ respectively defined in (11), (18) and (23).

3.4.4. Permutation condition. The last condition introduced by Lundgren is the permutation condition. The PDF has to be invariant under permutations of any pair of sample space-velocities and coordinates. This condition must also hold in the helical frame, yielding

$$\begin{aligned}&\tilde{f}_{(n)}\left(\tilde{\boldsymbol{v}}_{(1)},\ldots,\tilde{\boldsymbol{v}}_{(i)},\ldots,\tilde{\boldsymbol{v}}_{(j)},\ldots,\tilde{\boldsymbol{v}}_{(n)};\tilde{\boldsymbol{x}}_{(1)},\ldots,\tilde{\boldsymbol{x}}_{(i)},\ldots,\tilde{\boldsymbol{x}}_{(j)},\ldots,\tilde{\boldsymbol{x}}_{(n)}\right)\\&\quad=\tilde{f}_{(n)}\left(\tilde{\boldsymbol{v}}_{(1)},\ldots,\tilde{\boldsymbol{v}}_{(j)},\ldots,\tilde{\boldsymbol{v}}_{(i)},\ldots,\tilde{\boldsymbol{v}}_{(n)};\tilde{\boldsymbol{x}}_{(1)},\ldots,\tilde{\boldsymbol{x}}_{(j)},\ldots,\tilde{\boldsymbol{x}}_{(i)},\ldots,\boldsymbol{x}_{(n)}\right).\end{aligned} \tag{46}$$

3.4.5. Consistency condition. Ulinich and Lyubimov (1969) introduced the additional consistency condition, which was not mentioned in Lundgren (1967). It reflects the fact that $\boldsymbol{x}_{(i)}$ are not random variables but the discrete indices of the random variables $\boldsymbol{v}_{(i)}$. It relates the spatial-derivatives to the derivatives w.r.t. the sample space variables of higher order PDFs (Ulinich and Lyubimov 1969; see also Praturi *et al* 2020)

$$\frac{\partial f_{(n)}}{\partial\boldsymbol{x}_{(i)}}=-\frac{\partial}{\partial\boldsymbol{v}_{(i)}}\iint\frac{\partial f_{(n+1)}}{\partial\boldsymbol{x}_{(n+1)}}\boldsymbol{v}_{(n+1)}\delta\left(\boldsymbol{x}_{(n+1)}-\boldsymbol{x}_{(i)}\right)\mathrm{d}\boldsymbol{v}_{(n+1)}\mathrm{d}\boldsymbol{x}_{(n+1)}. \tag{47}$$

Applying the transformations gives

$$\frac{\partial\tilde{f}_{(n)}}{\partial\tilde{\boldsymbol{x}}_{(i)}}=-\frac{\partial}{\partial\tilde{\boldsymbol{v}}_{(i)}}\iint\frac{\partial\tilde{f}_{(n+1)}}{\partial\tilde{\boldsymbol{x}}_{(n+1)}}\tilde{\boldsymbol{v}}_{(n+1)}\delta\left(\tilde{\boldsymbol{x}}_{(n+1)}-\tilde{\boldsymbol{x}}_{(i)}\right)\frac{r_{(n+1)}}{2}\mathrm{d}\tilde{\boldsymbol{v}}_{(n+1)}\mathrm{d}\tilde{\boldsymbol{x}}_{(n+1)}. \tag{48}$$

3.5. Two-component helical flows

Lundgren's hierarchy can be further reduced by assuming a two-component flow, where the velocity component is $u^\eta = 0$. This case is interesting because it means that there is no more vortex stretching, i.e. similar to 2D turbulence. In this case the following approach must be chosen for the PDF to ensure that all moments, including u^η, generally disappear:

$$f_{(i)} = \frac{1}{J_{(i)}}\tilde{f}_{(i)}\left(v^r_{(1)}, v^\xi_{(1)}, \ldots, v^r_{(i)}, v^\xi_{(i)}; r_{(1)}, \xi_{(1)}, \ldots, r_{(i)}, \xi_{(i)}\right)\prod_{j=1}^{i}\delta\left(v^\eta_{(j)}\right), \tag{49}$$

Inserting the latter into the helically symmetric hierarchy (32), performing the integration over $v^\eta_{(n+1)}$ and then integrating the entire hierarchy with respect to all remaining sample space variables $v^\eta_{(1)}, \ldots, v^\eta_{(n)}$, gives the LMN hierarchy for a two-component helical flow. In this case the hierarchy is reduced w.r.t. v^η but otherwise looks identical to (32), i.e. v^η no longer appears in the equation.

4. Equations of the 1st statistical moment

As a simple check on the correctness of the above, we compare the Reynolds averaged helically symmetric Navier–Stokes (RANS) equation (50), obtained by simple averaging of Navier–Stokes, with the first moment transport equation obtained from the helically symmetric Lundgren hierarchy (32). To obtain the RANS equation, we apply the averaging operator $\langle\ldots\rangle$ to the helically symmetric Navier–Stokes equations (5), which gives the transport equation for the first order velocity moments. As with the original Navier–Stokes equations, these are independent of the coordinate η, but have three velocity components. The moment equations in the helically symmetric frame are finally

$$\begin{aligned}&\left\langle\frac{\partial u^r}{\partial t}\right\rangle + \left\langle u^r\frac{\partial u^r}{\partial r} + \frac{1}{B}u^\xi\frac{\partial u^r}{\partial \xi} - \frac{B^2}{r}\left(\frac{b}{r}u^\xi - au^\eta\right)^2\right\rangle \\ &\quad = -\left\langle\frac{\partial p}{\partial r}\right\rangle + \nu\left\langle\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u^r}{\partial r}\right) + \frac{1}{B^2}\frac{\partial^2 u^r}{\partial \xi^2} - \frac{1}{r^2}u^r + \frac{2bB}{r^2}\left(a\frac{\partial u^\eta}{\partial \xi} - \frac{b}{r}\frac{\partial u^\xi}{\partial \xi}\right)\right\rangle,\end{aligned} \tag{50a}$$

$$\begin{aligned}&\left\langle\frac{\partial u^\xi}{\partial t}\right\rangle + \left\langle u^r\frac{\partial u^\xi}{\partial r} + \frac{1}{B}u^\xi\frac{\partial u^\xi}{\partial \xi} - \frac{2abB^2}{r}u^r u^\eta + \frac{b^2B^2}{r^2}u^r u^\xi\right\rangle \\ &\quad = -\left\langle\frac{1}{B}\frac{\partial p}{\partial \xi}\right\rangle + \nu\left\langle\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u^\xi}{\partial r}\right) + \frac{1}{B^2}\frac{\partial^2 u^\xi}{\partial \xi^2} + \frac{a^4B^4-1}{r^2}u^\xi \\ &\qquad + \frac{2bB}{r}\left(\frac{b}{r^2}\frac{\partial u^r}{\partial \xi} - \frac{\partial \frac{aB}{r}u^\eta}{\partial r}\right)\right\rangle,\end{aligned} \tag{50b}$$

$$\begin{aligned}&\left\langle\frac{\partial u^\eta}{\partial t}\right\rangle + \left\langle u^r\frac{\partial u^\eta}{\partial r} + \frac{1}{B}u^\xi\frac{\partial u^\eta}{\partial \xi} + \frac{a^2B^2}{r}u^r u^\eta\right\rangle \\ &\quad = \nu\left\langle\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u^\eta}{\partial r}\right) + \frac{1}{B^2}\frac{\partial^2 u^\eta}{\partial \xi^2} + \frac{a^2B^2\left(a^2B^2-2\right)}{r^2}u^\eta - \frac{2abB}{r^2}\left(\frac{\partial u^r}{\partial \xi} - \frac{\partial Bu^\xi}{\partial r}\right)\right\rangle,\end{aligned} \tag{50c}$$

In a second step, we derive the 1st moment equations by multiplying the one-point LMN hierarchy, i.e. the equation (32) for $n = 1$ by $\tilde{\boldsymbol{v}}_{(1)}$ and integrating w.r.t. $\tilde{\boldsymbol{v}}_{(1)}$. The result must be identical to the moment equation (50) derived above by applying the averaging operator to the Navier–Stokes equation (5). To derive the velocity moments we use the definition $\langle\tilde{\boldsymbol{u}}_{(1)}\rangle = \int\tilde{\boldsymbol{v}}_{(1)}\tilde{f}_{(1)}\mathrm{d}\tilde{\boldsymbol{v}}_{(1)}$.

Deriving the one-point moment equation by $\int \tilde{\boldsymbol{v}}_{(1)}(32)|_{n=1}\mathrm{d}\tilde{\boldsymbol{v}}_{(1)}$ results in

$$\int \tilde{\boldsymbol{v}}_{(1)} \frac{\partial \tilde{f}_{(1)}}{\partial t}\mathrm{d}\tilde{\boldsymbol{v}}_{(1)} + \left(\boldsymbol{e}_{r_{(1)}} \frac{\partial}{\partial r_{(1)}} + \frac{1}{B_{(1)}}\boldsymbol{e}_{\xi_{(1)}} \frac{\partial}{\partial \xi_{(1)}}\right) \cdot \int \tilde{\boldsymbol{v}}_{(1)}\tilde{\boldsymbol{v}}_{(1)}\tilde{f}_{(1)}\mathrm{d}\tilde{\boldsymbol{v}}_{(1)}$$
$$+ \int \tilde{\boldsymbol{v}}_{(1)} \frac{\partial}{\partial \tilde{\boldsymbol{v}}_{(1)}} \cdot \left\{ -\frac{1}{4\pi} \int \left[\tilde{\boldsymbol{v}}_{(2)} \cdot \left(\boldsymbol{e}_{r_{(2)}} \frac{\partial}{\partial r_{(2)}} + \frac{1}{B_{(2)}}\boldsymbol{e}_{\xi_{(2)}} \frac{\partial}{\partial \xi_{(2)}}\right)\right]^2 \right.$$
$$\cdot \tilde{f}_{(2)} \mathcal{G}'\left(\tilde{\boldsymbol{x}}_{(1)}, \tilde{\boldsymbol{x}}_{(2)}\right) \frac{r_{(2)}}{2} dr_{(2)} \mathrm{d}\xi_{(2)} \mathrm{d}\tilde{\boldsymbol{v}}_{(2)}$$
$$\left. + \lim_{\tilde{\boldsymbol{x}}_{(2)} \to \tilde{\boldsymbol{x}}_{(1)}} \nu \left[\frac{\partial^2}{\partial r_{(2)}^2} + \frac{1}{r_{(2)}} \frac{\partial}{\partial r_{(2)}} + \frac{1}{B_{(2)}^2} \frac{\partial^2}{\partial \xi_{(2)}^2}\right] \int \tilde{\boldsymbol{v}}_{(2)}\tilde{f}_{(2)}\mathrm{d}\tilde{\boldsymbol{v}}_{(2)} \right\} \mathrm{d}\tilde{\boldsymbol{v}}_{(1)} = 0.$$

The goal is now to identify velocity moments in (4), which we do in the following for each term of the hierarchy successively.

4.1. Time derivative

We start with the time derivative term. Since temporal derivative and integration simply commute, the averaged velocity is easily identified by

$$\int \tilde{\boldsymbol{v}}_{(1)} \frac{\partial \tilde{f}_{(1)}}{\partial t}\mathrm{d}\tilde{\boldsymbol{v}}_{(1)} = \frac{\partial}{\partial t} \int \tilde{\boldsymbol{v}}_{(1)}\tilde{f}_{(1)}\mathrm{d}\tilde{\boldsymbol{v}}_{(1)} = \frac{\partial}{\partial t}\langle \tilde{\boldsymbol{u}}_{(1)} \rangle. \tag{51}$$

4.2. Convective term

The convective term is rewritten using (3)

$$\begin{aligned} \int \tilde{\boldsymbol{v}}_{(1)} \cdot \mathcal{C}\left(\tilde{\boldsymbol{v}}_{(1)}; \tilde{\boldsymbol{x}}_{(1)}\right) \mathrm{d}\tilde{\boldsymbol{v}}_{(1)} &= \tilde{\nabla}_{(1)} \cdot \int \tilde{\boldsymbol{v}}_{(1)}\tilde{\boldsymbol{v}}_{(1)}\tilde{f}_{(1)}\mathrm{d}\tilde{\boldsymbol{v}}_{(1)} \\ &= \tilde{\nabla}_{(1)} \cdot \langle \tilde{\boldsymbol{u}}_{(1)}\tilde{\boldsymbol{u}}_{(1)} \rangle. \end{aligned} \tag{52}$$

Further adjustment of the equation results in

$$\int \tilde{\boldsymbol{v}}_{(1)} \cdot \mathcal{C}\left(\tilde{\boldsymbol{v}}_{(1)}; \tilde{\boldsymbol{x}}_{(1)}\right) \mathrm{d}\tilde{\boldsymbol{v}}_{(1)} = \left\langle \left(\tilde{\boldsymbol{u}}_{(1)} \cdot \tilde{\nabla}_{(1)}\right) \tilde{\boldsymbol{u}}_{(1)} \right\rangle. \tag{53}$$

4.3. Viscous term

For the viscous term, we use symbolic notation and apply the reduction condition (36) or rather $\int \delta(\tilde{\boldsymbol{u}}_{(i)} - \tilde{\boldsymbol{v}}_{(i)})\mathrm{d}\tilde{\boldsymbol{v}}_{(i)} = 1$, which gives

$$\int \tilde{\boldsymbol{v}}_{(1)} \cdot \mathcal{V}\left(\tilde{\boldsymbol{v}}_{(1)}, \tilde{\boldsymbol{v}}_{(2)}; \tilde{\boldsymbol{x}}_{(1)}, \tilde{\boldsymbol{x}}_{(2)}\right) \mathrm{d}\tilde{\boldsymbol{v}}_{(1)} = \int \tilde{\boldsymbol{v}}_{(1)} \frac{\partial}{\partial \tilde{\boldsymbol{v}}_{(1)}} \cdot \lim_{\tilde{\boldsymbol{x}}_{(2)} \to \tilde{\boldsymbol{x}}_{(1)}} \nu \tilde{\Delta}_{(2)} \int \tilde{\boldsymbol{v}}_{(2)}\tilde{f}_{(2)}\mathrm{d}\tilde{\boldsymbol{v}}_{(2)}\mathrm{d}\tilde{\boldsymbol{v}}_{(1)}$$
$$= \int \tilde{\boldsymbol{v}}_{(1)} \frac{\partial}{\partial \tilde{\boldsymbol{v}}_{(1)}} \cdot \lim_{\tilde{\boldsymbol{x}}_{(2)} \to \tilde{\boldsymbol{x}}_{(1)}} \nu \tilde{\Delta}_{(2)} \int \langle \tilde{\boldsymbol{u}}_{(2)} \delta\left(\tilde{\boldsymbol{u}}_{(2)} - \tilde{\boldsymbol{v}}_{(2)}\right) \delta\left(\tilde{\boldsymbol{u}}_{(1)} - \tilde{\boldsymbol{v}}_{(1)}\right) \rangle \mathrm{d}\tilde{\boldsymbol{v}}_{(2)}\mathrm{d}\tilde{\boldsymbol{v}}_{(1)}$$
$$= \int \tilde{\boldsymbol{v}}_{(1)} \frac{\partial}{\partial \tilde{\boldsymbol{v}}_{(1)}} \cdot \lim_{\tilde{\boldsymbol{x}}_{(2)} \to \tilde{\boldsymbol{x}}_{(1)}} \nu \tilde{\Delta}_{(2)} \langle \tilde{\boldsymbol{u}}_{(2)} \delta\left(\tilde{\boldsymbol{u}}_{(1)} - \tilde{\boldsymbol{v}}_{(1)}\right) \rangle \mathrm{d}\tilde{\boldsymbol{v}}_{(1)}$$
$$= \int \tilde{\boldsymbol{v}}_{(1)} \frac{\partial}{\partial \tilde{\boldsymbol{v}}_{(1)}} \cdot \int \nu \tilde{\Delta}_{(2)} \langle \tilde{\boldsymbol{u}}_{(2)} \delta\left(\tilde{\boldsymbol{u}}_{(1)} - \tilde{\boldsymbol{v}}_{(1)}\right) \rangle \delta\left(\tilde{\boldsymbol{x}}_{(2)} - \tilde{\boldsymbol{x}}_{(1)}\right) \mathrm{d}\tilde{\boldsymbol{x}}_{(2)}\mathrm{d}\tilde{\boldsymbol{v}}_{(1)}$$

$$
\begin{aligned}
&= \int \tilde{\boldsymbol{v}}_{(1)} \frac{\partial}{\partial \tilde{\boldsymbol{v}}_{(1)}} \cdot \nu \tilde{\Delta}_{(1)} \langle \tilde{\boldsymbol{u}}_{(1)} \delta \left(\tilde{\boldsymbol{u}}_{(1)} - \tilde{\boldsymbol{v}}_{(1)} \right) \rangle \mathrm{d}\tilde{\boldsymbol{v}}_{(1)} \\
&= \nu \tilde{\Delta}_{(1)} \int \tilde{\boldsymbol{v}}_{(1)} \left\langle \frac{\partial}{\partial \tilde{\boldsymbol{v}}_{(1)}} \left(\tilde{\boldsymbol{u}}_{(1)} \delta \left(\tilde{\boldsymbol{u}}_{(1)} - \tilde{\boldsymbol{v}}_{(1)} \right) \right) \right\rangle \mathrm{d}\tilde{\boldsymbol{v}}_{(1)} = -\nu \tilde{\Delta}_{(1)} \int \tilde{\boldsymbol{v}}_{(1)} \langle \delta \left(\tilde{\boldsymbol{u}}_{(1)} - \tilde{\boldsymbol{v}}_{(1)} \right) \rangle \mathrm{d}\tilde{\boldsymbol{v}}_{(1)} \\
&= -\nu \tilde{\Delta}_{(1)} \int \tilde{\boldsymbol{v}}_{(1)} \tilde{f}_{(1)} \mathrm{d}\tilde{\boldsymbol{v}}_{(1)} = -\nu \tilde{\Delta}_{(1)} \langle \tilde{\boldsymbol{u}}_{(1)} \rangle. \qquad (54)
\end{aligned}
$$

4.4. Pressure term

Lastly, the pressure term is given by

$$
\begin{aligned}
&\int \tilde{\boldsymbol{v}}_{(1)} \cdot \mathcal{P} \left(\tilde{\boldsymbol{v}}_{(1)}, \tilde{\boldsymbol{v}}_{(2)}; \tilde{\boldsymbol{x}}_{(1)}, \tilde{\boldsymbol{x}}_{(2)} \right) \mathrm{d}\tilde{\boldsymbol{v}}_{(1)} \\
&\quad = -\int \tilde{\boldsymbol{v}}_{(1)} \frac{\partial}{\partial \tilde{\boldsymbol{v}}_{(1)}} \cdot \frac{1}{4\pi} \int \left[\tilde{\boldsymbol{v}}_{(2)} \cdot \left(\boldsymbol{e}_{r_{(2)}} \frac{\partial}{\partial r_{(2)}} + \frac{1}{B_{(2)}} \boldsymbol{e}_{\xi_{(2)}} \frac{\partial}{\partial \xi_{(2)}} \right) \right]^2 \\
&\qquad \cdot \tilde{f} \mathcal{G}' \left(\tilde{\boldsymbol{x}}_{(1)}, \tilde{\boldsymbol{x}}_{(2)} \right) \frac{r_{(2)}}{2} \mathrm{d}r_{(2)} \mathrm{d}\xi_{(2)} \mathrm{d}\tilde{\boldsymbol{v}}_{(2)} \mathrm{d}\tilde{\boldsymbol{v}}_{(1)}. \qquad (55)
\end{aligned}
$$

Using partial integration w.r.t. $\tilde{\boldsymbol{v}}_{(1)}$ leads to

$$
\begin{aligned}
&\int \tilde{\boldsymbol{v}}_{(1)} \cdot \mathcal{P} \left(\tilde{\boldsymbol{v}}_{(1)}, \tilde{\boldsymbol{v}}_{(2)}; \tilde{\boldsymbol{x}}_{(1)}, \tilde{\boldsymbol{x}}_{(2)} \right) \mathrm{d}\tilde{\boldsymbol{v}}_{(1)} \\
&\quad = \frac{1}{4\pi} \int \left[\tilde{\boldsymbol{v}}_{(2)} \cdot \tilde{\nabla}_{(2)} \right]^2 \tilde{f}_{(2)} \mathcal{G}' \left(\boldsymbol{x}_{(1)}, \boldsymbol{x}_{(2)} \right) \frac{r_{(2)}}{2} \mathrm{d}r_{(2)} \mathrm{d}\xi_{(2)} \mathrm{d}\tilde{\boldsymbol{v}}_{(2)}. \qquad (56)
\end{aligned}
$$

Now the pressure is identified using Green's function

$$
\frac{1}{4\pi} \int \left[\tilde{\boldsymbol{v}}_{(2)} \cdot \tilde{\nabla}_{(2)} \right]^2 \tilde{f}_{(2)} \tilde{\nabla}_{(1)} \left(|\tilde{\boldsymbol{x}}_{(1)} - \tilde{\boldsymbol{x}}_{(2)}| \right)^{-1} \frac{r_{(2)}}{2} \mathrm{d}\tilde{\boldsymbol{x}}_{(2)} \mathrm{d}\tilde{\boldsymbol{v}}_{(2)} = \tilde{\nabla}_{(1)} \tilde{p}. \qquad (57)
$$

Collecting the terms (51), (53), (54) and (57) the 1st moment of the LMN hierarchy then reads

$$
\frac{\partial}{\partial t} \langle \tilde{\boldsymbol{u}}_{(1)} \rangle + \left\langle \left(\tilde{\boldsymbol{u}}_{(1)} \cdot \tilde{\nabla}_{(1)} \right) \tilde{\boldsymbol{u}}_{(1)} \right\rangle = -\tilde{\nabla}_{(1)} \langle \tilde{p}_{(1)} \rangle + \nu \tilde{\Delta}_{(1)} \langle \tilde{\boldsymbol{u}}_{(1)} \rangle, \qquad (58)
$$

which is equal to the averaged Navier–Stokes equations (5). Inserting the transformed derivatives and velocities $\tilde{\boldsymbol{u}} = u^r \boldsymbol{e}_r + u^\xi \boldsymbol{e}_\xi + u^\eta \boldsymbol{e}_\eta$ into this equation yields (50).

5. Characteristic function representation of helical turbulence

Characteristic functions provide an alternative route to moment equations. They are derived by Fourier transformation of the PDF in sample space and have the key advantage over the PDF-description that velocity moments follow from differentiation, not from integration as it is the case for PDFs. The characteristic function is a scalar density, defined by

$$
\varphi_{(n)} \left(\boldsymbol{x}_{(1)}, \boldsymbol{\kappa}_{(1)} \ldots \boldsymbol{x}_{(n)}, \boldsymbol{\kappa}_{(n)} \right) = \int f_{(n)} \left(V; X \right) e^{\mathrm{i} \sum_{j=1}^{n} \boldsymbol{\kappa}_{(j)} \cdot \boldsymbol{v}_{(j)}} \mathrm{d}V. \qquad (59)
$$

Corresponding k^{th} order statistical moments $m^{(k)}$, with k being a natural number are given by

$$m^{(k)} = \frac{1}{\mathrm{i}^k} \frac{\partial^k \varphi(\boldsymbol{x}, \boldsymbol{\kappa})}{\partial \boldsymbol{\kappa}^k}\bigg|_{\boldsymbol{\kappa}=0}. \tag{60}$$

The vector $\boldsymbol{\kappa}$ is transformed the same way as the sample space velocity $\boldsymbol{v}$ (see appendix A.1)

$$\tilde{\boldsymbol{\kappa}} = \kappa^r \boldsymbol{e_r} + \kappa^\xi \boldsymbol{e}_\xi + \kappa^\eta \boldsymbol{e}_\eta, \tag{61}$$

$$\varphi_{(n)}\left(\boldsymbol{x}_{(1)}, \boldsymbol{\kappa}_{(1)} \ldots \boldsymbol{x}_{(n)}, \boldsymbol{\kappa}_{(n)}\right) = \tilde{\varphi}_{(n)}\left(\tilde{\boldsymbol{x}}_{(1)}, \tilde{\boldsymbol{\kappa}}_{(1)} \ldots \tilde{\boldsymbol{x}}_{(n)}, \tilde{\boldsymbol{\kappa}}_{(n)}\right), \tag{62}$$

and the $\tilde{\boldsymbol{\kappa}}$-derivative reads

$$\frac{\partial}{\partial \tilde{\boldsymbol{\kappa}}} = \frac{\partial}{\partial \kappa^r}\boldsymbol{e_r} + \frac{\partial}{\partial \kappa^\xi}\boldsymbol{e}_\xi + \frac{\partial}{\partial \kappa^\eta}\boldsymbol{e}_\eta. \tag{63}$$

From here on we switch to the helically symmetric frame. The hierarchy for the characteristic function is derived from the LMN hierarchy (32) by multiplying it with $\mathrm{e}^{\mathrm{i}\sum_{j=1}^{n} \tilde{\boldsymbol{\kappa}}_{(j)} \cdot \tilde{\boldsymbol{v}}_{(j)}}$ and integrating everything w.r.t. $\tilde{\boldsymbol{v}}_{(j)}$, where $j = 1 \ldots n$. The hierarchy for the characteristic functions then reads

$$\begin{aligned} 0 = &\left[\frac{\partial}{\partial t} - \mathrm{i}\sum_{j=1}^{n} \tilde{\nabla}_{(j)} \frac{\partial}{\partial \tilde{\boldsymbol{\kappa}}_{(j)}}\right]\tilde{\varphi}_{(n)} + \sum_{j=1}^{n} \tilde{\boldsymbol{\kappa}}_{(j)} \cdot \left[\int \mathrm{i}\frac{1}{4\pi}\mathcal{G}'\left(\tilde{\boldsymbol{x}}_{(j)}, \tilde{\boldsymbol{x}}_{(n+1)}\right)\right. \\ &\times \left(\tilde{\nabla}_{(n+1)} \otimes \tilde{\nabla}_{(n+1)}\right) : \left(\frac{\partial}{\partial \tilde{\boldsymbol{\kappa}}_{(n+1)}} \otimes \frac{\partial \tilde{\varphi}_{(n+1)}}{\partial \tilde{\boldsymbol{\kappa}}_{(n+1)}}\right)\bigg|_{\tilde{\boldsymbol{\kappa}}_{(n+1)}=0} \frac{r_{(n+1)}}{2} \mathrm{d}r_{(n+1)} \mathrm{d}\xi_{(n+1)} \\ &\left. - \lim_{\tilde{\boldsymbol{x}}_{(n+1)} \to \tilde{\boldsymbol{x}}_{(j)}} \nu \tilde{\nabla}^2_{(n+1)} \frac{\partial \tilde{\varphi}_{(n+1)}}{\partial \tilde{\boldsymbol{\kappa}}_{(n+1)}}\bigg|_{\tilde{\boldsymbol{\kappa}}_{(n+1)}=0}\right]. \end{aligned} \tag{64}$$

In order to derive this equation in a helical frame, the inner part of the pressure term was rewritten as

$$\begin{aligned} &\int \left(\tilde{\boldsymbol{v}}_{(n+1)} \cdot \tilde{\nabla}_{(n+1)}\right)^2 \tilde{f}_{(n+1)} e^{\mathrm{i}\sum_{j=1}^{n}\left(\tilde{\boldsymbol{\kappa}}_{(j)} \cdot \tilde{\boldsymbol{v}}_{(j)}\right)} \mathrm{d}\tilde{\boldsymbol{v}}_{(1)} \ldots \mathrm{d}\tilde{\boldsymbol{v}}_{(n+1)} \\ &\quad = -\left(\tilde{\nabla}_{(n+1)} \otimes \tilde{\nabla}_{(n+1)}\right) : \left(\frac{\partial}{\partial \tilde{\boldsymbol{\kappa}}_{(n+1)}} \otimes \frac{\partial \tilde{\varphi}_{(n+1)}}{\partial \tilde{\boldsymbol{\kappa}}_{(n+1)}}\right)\bigg|_{\tilde{\boldsymbol{\kappa}}_{(n+1)}=0}. \end{aligned} \tag{65}$$

5.1. Side conditions in characteristic function form

The respective side conditions, which are derived by carrying out a Fourier-transformation onto them will be given subsequently. The normalization condition (33) yields

$$\int \tilde{f}_{(n)} \mathrm{d}\tilde{V}_{(n)} = 1 \;\Rightarrow\; \tilde{\varphi}_{(n)}\bigg|_{\tilde{\boldsymbol{\kappa}}_{(1)} \ldots \tilde{\boldsymbol{\kappa}}_{(n)}=0} = 1. \tag{66}$$

The reduction condition (34) is transformed to

$$\int \tilde{f}_{(n+1)} \mathrm{d}\tilde{\boldsymbol{v}}_{(n+1)} = \tilde{f}_{(n)} \;\Rightarrow\; \tilde{\varphi}_{(n+1)}\bigg|_{\tilde{\boldsymbol{\kappa}}_{(n+1)}=0} = \tilde{\varphi}_{(n)}. \tag{67}$$

Further, the coincidence condition (38) is Fourier transformed as well, where we made use of the delta-sifting property, leading to

$$\lim_{\tilde{\boldsymbol{x}}_{(n+1)}\to\tilde{\boldsymbol{x}}_{(n)}} \tilde{\varphi}_{(n+1)}\left(\tilde{\boldsymbol{x}}_{(1)},\tilde{\boldsymbol{\kappa}}_{(1)},\ldots,\tilde{\boldsymbol{x}}_{(n+1)},\tilde{\boldsymbol{\kappa}}_{(n+1)}\right) = \tilde{\varphi}_{(n)}\left(\tilde{\boldsymbol{x}}_{(1)},\tilde{\boldsymbol{\kappa}}_{(1)},\ldots,\tilde{\boldsymbol{x}}_{(n)},\tilde{\boldsymbol{\kappa}}_{(n)}+\tilde{\boldsymbol{\kappa}}_{(n+1)}\right). \tag{68}$$

The divergence condition (45) after Fourier transformation for $n \geqslant k$ and helically symmetric turbulence, i.e. $\frac{\partial}{\partial\eta}=0$, reads with operation (23)

$$\tilde{\nabla}_{(k)}\cdot\frac{\partial}{\partial\tilde{\boldsymbol{\kappa}}_{(k)}}\tilde{\varphi}_{(n)}\bigg|_{\tilde{\boldsymbol{\kappa}}_{(k)}=0}=0. \tag{69}$$

In general, the Fourier transform commutes with the permutation of the arguments $\tilde{\boldsymbol{\kappa}}_{(i)}$ and $\tilde{\boldsymbol{x}}_{(i)}$. The permutation condition (46) therefore transforms to

$$\tilde{\varphi}_{(n)}\left(\tilde{\boldsymbol{\kappa}}_{(1)},\tilde{\boldsymbol{x}}_{(1)},\ldots,\tilde{\boldsymbol{\kappa}}_{(i)},\tilde{\boldsymbol{x}}_{(i)},\ldots,\tilde{\boldsymbol{\kappa}}_{(j)},\tilde{\boldsymbol{x}}_{(j)},\ldots,\tilde{\boldsymbol{\kappa}}_{(n)},\tilde{\boldsymbol{x}}_{(n)}\right) = \tilde{\varphi}_{(n)}\left(\tilde{\boldsymbol{\kappa}}_{(1)},\tilde{\boldsymbol{x}}_{(1)},\ldots,\tilde{\boldsymbol{\kappa}}_{(j)},\tilde{\boldsymbol{x}}_{(j)},\ldots,\tilde{\boldsymbol{\kappa}}_{(i)},\tilde{\boldsymbol{x}}_{(i)},\ldots,\tilde{\boldsymbol{\kappa}}_{(n)},\tilde{\boldsymbol{x}}_{(n)}\right). \tag{70}$$

Lastly, Fourier transformation of the consistency condition (48) yields

$$\frac{\partial\tilde{\varphi}_{(n)}}{\partial\tilde{\boldsymbol{x}}_{(i)}} = -\int\left(\left(\tilde{\boldsymbol{\kappa}}_{(i)}\cdot\frac{\partial}{\partial\tilde{\boldsymbol{\kappa}}_{(n+1)}}\right)\frac{\partial\tilde{\varphi}_{(n+1)}}{\partial\tilde{\boldsymbol{x}}_{(n+1)}}\right)\bigg|_{\tilde{\boldsymbol{\kappa}}_{(n+1)}=0}\delta\left(\tilde{\boldsymbol{x}}_{(n+1)}-\tilde{\boldsymbol{x}}_{(i)}\right)\frac{r_{(n+1)}}{2}\mathrm{d}\tilde{\boldsymbol{x}}_{(n+1)}. \tag{71}$$

6. Conclusion and outlook

In this paper, incompressible helically symmetric turbulence has been considered in a suitable helical coordinate system. The LMN hierarchy and the associated side conditions have been derived for helical coordinates, which are dimensionally reduced compared to Lundgren's hierarchy in Cartesian coordinates. In order to verify these results, the moment equations derived from the LMN hierarchy for helically symmetric turbulence were compared with the 1st moment equations derived from the averaged Navier–Stokes equations in helical coordinates leading to identical systems. An alternative way to an infinite hierarchy of moment equations is shown by deriving the hierarchy and the corresponding side conditions for characteristic functions using helical coordinates.

On the basis of the present results, it will probably be difficult to find closure approaches for the LMN hierarchy for helically symmetric flows, or it will be very complicated to justify them well from a physical-mathematical point of view. We are therefore planning two alternative approaches: (i) In an ongoing research project, the helically reduced LMN hierarchy is compared with the corresponding DNS in a helically symmetric system. The plan is to extract corresponding PDF from the DNS data. (ii) We were able to perform a spectral-like decomposition for the general LMN hierarchy, which led to a non-linear eigenvalue equations. The assumption is that this will also be possible for the helically symmetric LMN. One of the key questions to be answered here is that of vortex stretching and the anomalous scaling of dissipation according to Kolmolgorov. Depending on the velocity, i.e. if $u^{\eta}\neq 0$ applies, vortex stretching occurs similar to 3D turbulence and for the case $u^{\eta}=0$ there is no vortex stretching similar to 2D turbulence. The comparison of helical turbulence with classical 2D and 3D turbulence is expected to lead to a deeper understanding of the transition between the two cases.

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Appendix

A.1. Transformation of sample space variable and derivative

In order to transform the sample space variable in Cartesian coordinates to those in helical coordinates, we need to define the sample space variables in helical coordinates. This is done using the rotation matrix $\boldsymbol{S}$, were $\theta = -\frac{ab\eta-\xi}{2b}$

$$(\boldsymbol{e_x}, \boldsymbol{e_y}, \boldsymbol{e_z}) = (\boldsymbol{e_r}, \boldsymbol{e_\xi}, \boldsymbol{e_\eta}) \cdot \boldsymbol{S}^T \qquad \boldsymbol{S} = \begin{pmatrix} \cos\theta & -\frac{bB\sin\theta}{r} & aB\sin\theta \\ \sin\theta & \frac{bB\cos\theta}{r} & -aB\cos\theta \\ 0 & aB & \frac{bB}{r} \end{pmatrix}. \tag{A.1}$$

The Cartesian sample space variable components are defined by

$$\begin{aligned} v^x &= v^r\cos\theta - v^\xi\frac{bB}{r}\sin\theta + v^\eta aB\sin\theta, \\ v^y &= v^r\sin\theta + v^\xi\frac{bB}{r}\cos\theta - v^\eta aB\cos\theta, \\ v^z &= v^\xi aB + v^\eta\frac{bB}{r}. \end{aligned} \tag{A.2}$$

Inserting the definition of the basis vectors

$$\begin{aligned} \boldsymbol{e_x} &= \cos\theta\boldsymbol{e_r} - \frac{bB\sin\theta}{r}\boldsymbol{e}_\xi + aB\sin\theta\boldsymbol{e}_\eta, \\ \boldsymbol{e_y} &= \sin\theta\boldsymbol{e_r} + \frac{bB\cos\theta}{r}\boldsymbol{e}_\xi - aB\cos\theta\boldsymbol{e}_\eta, \\ \boldsymbol{e_z} &= aB\boldsymbol{e}_\xi + \frac{bB}{r}\boldsymbol{e}_\eta. \end{aligned} \tag{A.3}$$

as well as (A.2) into the definition of the sample space variable $\boldsymbol{v} = v^x\boldsymbol{e_x} + v^y\boldsymbol{e_y} + v^y\boldsymbol{e_y}$, yields the sample space variable in helical coordinates (A.4)

$$\tilde{\boldsymbol{v}} = v^r\boldsymbol{e_r} + v^\xi\boldsymbol{e}_\xi + v^\eta\boldsymbol{e}_\eta. \tag{A.4}$$

The derivative w.r.t. $\boldsymbol{v}$ is transformed the following way

$$\begin{aligned}\frac{\partial}{\partial \boldsymbol{v}} &= \frac{\partial}{\partial v^x}\boldsymbol{e_x} + \frac{\partial}{\partial v^y}\boldsymbol{e_y} + \frac{\partial}{\partial v^z}\boldsymbol{e_z} \\ &= \boldsymbol{e_x}\left[\frac{\partial v^r}{\partial v^x}\frac{\partial}{\partial v^r} + \frac{\partial v^\xi}{\partial v^x}\frac{\partial}{\partial v^\xi} + \frac{\partial_{(\eta)}}{\partial v^x}\frac{\partial}{\partial_{(\eta)}}\right] + \boldsymbol{e_y}\left[\frac{\partial v^r}{\partial v^y}\frac{\partial}{\partial v^r} + \frac{\partial v^\xi}{\partial v^y}\frac{\partial}{\partial v^\xi} + \frac{\partial_{(\eta)}}{\partial v^y}\frac{\partial}{\partial_{(\eta)}}\right] \\ &\quad + \boldsymbol{e_z}\left[\frac{\partial v^r}{\partial v^z}\frac{\partial}{\partial v^r} + \frac{\partial v^\xi}{\partial v^z}\frac{\partial}{\partial v^\xi} + \frac{\partial_{(\eta)}}{\partial v^z}\frac{\partial}{\partial_{(\eta)}}\right].\end{aligned} \tag{A.5}$$

The Cartesian basis vectors are expressed using the helical basis vectors, which results in

$$\frac{\partial}{\partial \tilde{\boldsymbol{v}}} = \frac{\partial}{\partial v^r}\boldsymbol{e_r} + \frac{\partial}{\partial v^\xi}\boldsymbol{e}_\xi + \frac{\partial}{\partial v^\eta}\boldsymbol{e}_\eta, \tag{A.6}$$

and this final result is given in equation (22).

A.2. Laplacian in helical coordinates

The Laplace operator is the second order differential operator and defined as the divergence of the gradient. It is therefore calculated using the nabla-operator as $\Delta f = \nabla^2 f = \nabla \cdot \nabla f$, where f is a twice-differentiable function. In order to derive the Laplace operator the derivatives of the base vectors are used

$$\begin{aligned}&\frac{\partial \boldsymbol{e_r}}{\partial r} = 0, \qquad \frac{\partial \boldsymbol{e_r}}{\partial \xi} = -\frac{\sin\theta}{2b}\boldsymbol{e_x} + \frac{\cos\theta}{2b}\boldsymbol{e_y}, \qquad \frac{\partial \boldsymbol{e_r}}{\partial \eta} = \frac{a\sin\theta}{2}\boldsymbol{e_x} - \frac{a\cos\theta}{2}\boldsymbol{e_y}, \\ &\frac{\partial \boldsymbol{e}_\xi}{\partial r} = \frac{ab}{a^2r^2+b^2}\boldsymbol{e}_\eta, \qquad \frac{\partial \boldsymbol{e}_\xi}{\partial \xi} = -\frac{1}{2\sqrt{a^2r^2+b^2}}\boldsymbol{e_r}, \qquad \frac{\partial \boldsymbol{e}_\xi}{\partial \eta} = \frac{ab}{2\sqrt{a^2r^2+b^2}}\boldsymbol{e_r}, \\ &\frac{\partial \boldsymbol{e}_\eta}{\partial r} = -\frac{ab}{a^2r^2+b^2}\boldsymbol{e}_\xi, \qquad \frac{\partial \boldsymbol{e}_\eta}{\partial \xi} = \frac{ar}{2b\sqrt{a^2r^2+b^2}}\boldsymbol{e_r}, \qquad \frac{\partial \boldsymbol{e}_\eta}{\partial \eta} = -\frac{a^2r}{2\sqrt{a^2r^2+b^2}}\boldsymbol{e_r}.\end{aligned}$$

For an orthonormal coordinate system

$$\boldsymbol{e_i}\boldsymbol{e_j} = \delta_{ij} \tag{A.7}$$

holds. The Laplace-operator for helical coordinates then reads

$$\begin{aligned}\tilde{\Delta} = \tilde{\nabla}^2 &= \frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{B^2}\frac{\partial^2}{\partial \xi^2} + \frac{2\left(a^2r^2-b^2\right)}{bar^2}\frac{\partial^2}{\partial\xi\,\partial\eta} + \frac{4B^2}{r^2}\frac{\partial^2}{\partial\eta^2} \\ &\quad + \left(\frac{B\left(a^2r^2-b^2\right)}{bar^2}\right)^2\frac{\partial^2}{\partial\eta^2}.\end{aligned} \tag{A.8}$$

Which, for helically symmetric flows, where $\frac{\partial}{\partial\eta} = 0$, results in

$$\tilde{\Delta} = \frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{B^2}\frac{\partial^2}{\partial\xi^2}. \tag{A.9}$$

A.3. Further calculations for Green's term

Equation (30) can be derived with respect to $\tilde{\boldsymbol{x}}_{(j)}$ assuming $\frac{\partial}{\partial \eta} = 0$ yielding

$$\mathcal{G}' = \frac{4}{ab}\begin{bmatrix} \dfrac{\left(\frac{\partial\left[(\xi_I-\pi b)^2\right]}{\partial r_{(j)}}+\frac{\partial\left[a^2 r_I^2\right]}{\partial r_{(j)}}\right)}{2\sqrt{(\xi_I-\pi b)^2+a^2 r_I^2}} - \dfrac{\left(\frac{\partial\left[(\xi_I-b)^2\right]}{\partial r_{(j)}}+\frac{\partial\left[a^2 r_I^2\right]}{\partial r_{(j)}}\right)}{2\sqrt{(\xi_I-b)^2+a^2 r_I^2}} \\ \dfrac{2\left(\sqrt{(\xi_I-\pi b)^2+a^2 r_I^2}-\sqrt{(\xi_I-b)^2+a^2 r_I^2}\right)}{a} + \dfrac{2(1-\pi)b}{a} \end{bmatrix} \boldsymbol{e}_{r_{(j)}}$$
$$+ \frac{4}{abB_{(j)}}\begin{bmatrix} \dfrac{\left(\frac{\partial\left[(\xi_I-\pi b)^2\right]}{\partial \xi_{(j)}}+\frac{\partial\left[a^2 r_I^2\right]}{\partial \xi_{(j)}}\right)}{2\sqrt{(\xi_I-\pi b)^2+a^2 r_I^2}} - \dfrac{\left(\frac{\partial\left[(\xi_I-b)^2\right]}{\partial \xi_{(j)}}+\frac{\partial\left[a^2 r_I^2\right]}{\partial \xi_{(j)}}\right)}{2\sqrt{(\xi_I-b)^2+a^2 r_I^2}} \\ \dfrac{2\left(\sqrt{(\xi_I-b)^2+a^2 r_I^2}-\sqrt{(\xi_I-b)^2+a^2 r_I^2}\right)}{a} + \dfrac{2(1-\pi)b}{a} \end{bmatrix} \boldsymbol{e}_{\xi_{(j)}}. \tag{A.10}$$

The derivatives $\frac{\partial \xi_I}{\partial r_{(j)}}, \frac{\partial \xi_I}{\partial \xi_{(j)}}, \frac{\partial r_I}{\partial r_{(j)}}, \frac{\partial r_I}{\partial \xi_{(j)}}$ will in the following be derived as solutions of a system of linear equations. The increment vectors are implicitly defined by

$$\boldsymbol{F} = 0 = \boldsymbol{x}_I + \left(\boldsymbol{x}_{(j)} - \boldsymbol{x}_{(n+1)}\right) = \begin{pmatrix} F_1 \\ F_2 \\ F_3 \end{pmatrix} = 0. \tag{A.11}$$

From this vector-valued function derivatives are taken w.r.t the increment coordinates. Exemplary derivatives w.r.t. $\xi_(j)$ gives:

$$0 = \frac{\partial \boldsymbol{F}}{\partial \xi_{(j)}} = \begin{pmatrix} \frac{\partial \xi_I}{\partial \xi_{(j)}} \cdot \frac{\partial F_1}{\partial \xi_I} + \frac{\partial r_I}{\partial \xi_{(j)}} \cdot \frac{\partial F_1}{\partial r_I} + \frac{\partial \eta_I}{\partial \xi_{(j)}} \cdot \frac{\partial F_1}{\partial \eta_I} + \frac{\partial F_1}{\partial \xi_{(j)}} \\ \frac{\partial \xi_I}{\partial \xi_{(j)}} \cdot \frac{\partial F_2}{\partial \xi_I} + \frac{\partial r_I}{\partial \xi_{(j)}} \cdot \frac{\partial F_2}{\partial r_I} + \frac{\partial \eta_I}{\partial \xi_{(j)}} \cdot \frac{\partial F_2}{\partial \eta_I} + \frac{\partial F_2}{\partial \xi_{(j)}} \\ \frac{\partial \xi_I}{\partial \xi_{(j)}} \cdot \frac{\partial F_3}{\partial \xi_I} + \frac{\partial r_I}{\partial \xi_{(j)}} \cdot \frac{\partial F_3}{\partial r_I} + \frac{\partial \eta_I}{\partial \xi_{(j)}} \cdot \frac{\partial F_3}{\partial \eta_I} + \frac{\partial F_3}{\partial \xi_{(j)}} \end{pmatrix} \tag{A.12}$$

where all derivatives $\frac{\partial F_i}{\partial \xi_I}, \frac{\partial F_i}{\partial r_I}, \frac{\partial F_i}{\partial \eta_I}, \frac{\partial F_i}{\partial \xi_{(j)}}$ are known, since they follow from the definition of an increment vector w.r.t a Cartesian basis, given by (appendix A.3). An analogous procedure is also done by calculating $\frac{\partial \boldsymbol{F}}{\partial r_{(j)}}$ in the same way.

The resulting system describes a set of linear equations for the unknown derivatives of increment coordinates w.r.t. the absolute coordinates, occurring in (30). The system is then solved to:

$$\frac{\partial \xi_I}{\partial r_{(j)}} = -\left[b r_I \left(|\cos(\theta_I)|^2 \sin^2\left(\theta_{(j)}\right) sgn\left(\cos\left(\theta_{(j)}\right)\right) - \sin^2(\theta_I)\,|\cos\left(\theta_{(j)}\right)|^2\right)\right] /$$
$$\times \left[\sin(\theta_I)\, r_I^2 \left(\sin^2(\theta_I)\, sgn\left(\cos(\theta_I)\right) + |\cos(\theta_I)|\cos(\theta_I)\right)|\cos(\theta_I)|\right] \tag{A.13}$$

$$\frac{\partial r_I}{\partial r_{(j)}} = \left[r_I \left(|\cos(\theta_I)| \sin^2(\theta_1) sgn(\cos(\theta_{(j)})) + sgn(\cos(\theta_I)) \sin^2(\theta_1)\right.\right.$$
$$\left.\left. - sgn(\cos(\theta_I))|\cos(\theta_{(j)})|^2 \cos(\theta_I)\right)\right] /$$
$$\times \left[|\cos(\theta_I)|\, r_I \left(\sin^2(\theta_I)\, sgn\left(\cos(\theta_I)\right) + |\cos(\theta_I)|\cos(\theta_I)\right)\right] \tag{A.14}$$

$$\frac{\partial \eta_I}{\partial r_{(j)}} = \left[r_I\left(|\cos(\theta_I)|^2 \sin^2\left(\theta_{(j)}\right) sgn\left(cos\left(\theta_{(j)}\right)\right) - \sin^2(\theta_I)\,|\cos\left(\theta_{(j)}\right)|^2\right)\right] / \\ \times \left[\sin(\theta_I)\, a r_I^2 \left(\sin^2(\theta_I)\, sgn(\cos(\theta_I)) + |\cos(\theta_I)|\cos(\theta_I)\right)|\cos(\theta_I)|\right] \tag{A.15}$$

$$\frac{\partial \xi_I}{\partial \xi_{(j)}} = [0.5(|\cos(\theta_I)|\sin^3(\theta_I) sgn(cos(\theta_I)) r_I^2 \\ + \sin^2(\theta_I)\,|\cos\left(\theta_{(j)}\right)|\sin\left(\theta_{(j)}\right) sgn\left(\cos\left(\theta_{(j)}\right)\right) r_{(j)} + \cos(\theta_I)\sin(\theta_I)\, r_I^2 |\cos(\theta_I)|^2 \\ + \cos(\theta_{(j)})\sin(\theta_{(j)}) r_{(j)}^2 |\cos(\theta_I)|^2)] / \\ \times \left[\sin(\theta_I)\, r_I^2 \left(\sin^2(\theta_I)\, sgn(\cos(\theta_I)) + |\cos(\theta_I)|\cos(\theta_I)\right)|\cos(\theta_I)|\right] \tag{A.16}$$

$$\frac{\partial r_I}{\partial \xi_{(j)}} = -[0.5 r_I^2 \sin(\theta_{(j)}) sgn(cos(\theta_{(j)}))(sgn(cos(\theta_{(j)})) cos(\theta_{(j)}) |\cos(\theta_I)| \\ - sgn(\cos(\theta_I))|\cos(\theta_I)|\cos(\theta_I))] / \\ \times \left[|\cos(\theta_I)|\, r_I sgn(cos(\theta_I))\, b\left(\sin^2(\theta_I)\, sgn(\cos(\theta_I)) + |\cos(\theta_I)|\cos(\theta_I)\right)\right]$$

$$\frac{\partial \eta_I}{\partial \xi_{(j)}} = [0.5(|\cos(\theta_I)|\sin^3(\theta_I) sgn(cos(\theta_I)) r_I^2 \\ - \sin^2(\theta_I)\,|\cos\left(\theta_{(j)}\right)|\sin\left(\theta_{(j)}\right) sgn\left(\cos\left(\theta_{(j)}\right)\right) r_{(j)}^2 \\ + \cos(\theta_I)\sin(\theta_I) r_I^2 |\cos(\theta_I)|^2 - sgn(\cos(\theta_{(j)}))\cos(\theta_{(j)})\sin(\theta_{(j)}) r_{(j)}^2 |\cos(\theta_I)|^2)] / \\ \times \left[\sin(\theta_I)\, a b r_I^2 \left(\sin^2(\theta_I)\, sgn(\cos(\theta_I)) + |\cos(\theta_I)|\cos(\theta_I)\right)|\cos(\theta_I)|\right]. \tag{A.17}$$

Note that is has been assumed that $cos(\theta_k) \neq 0$ in order to calculate the derivative of the signum functions.

The increment coordinates in the latter equations are specified by solving (appendix A.3), explicitly given by

$$\pm r_I |\cos(\theta_I)| = \pm r_{(n+1)} |\cos\left(\theta_{(n+1)}\right)| \mp r_{(j)} |\cos\left(\theta_{(j)}\right)| \tag{A.18}$$

$$\pm r_I \sin(\theta_I)\, sgn(\cos(\theta_I)) = \pm r_{(n+1)} \sin\left(\theta_{(n+1)}\right) sgn\left(\cos\left(\theta_{(n+1)}\right)\right) \\ \mp r_{(j)} \sin\left(\theta_{(j)}\right) sgn\left(\cos\left(\theta_{(j)}\right)\right) \tag{A.19}$$

$$\frac{1}{2}\left(\frac{\xi_I}{a} + b\eta_I\right) = \frac{1}{2}\left(\frac{\xi_{(n+1)} - \xi_{(j)}}{a} + b\left(\eta_{(n+1)} - \eta_{(j)}\right)\right)$$

which would together with (A.10) provide the specific solution of Green's term. However, solving this system must be done numerically for the specific case.

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